
Open Problem (Inverse Galois Problem)

Whether or not every finite group G appears as the Galois group of some Galois extension of the rational number $\mathbb{Q}$. i.e.

$$\text{Gal}(K/\mathbb{Q}) = G$$

Equivalently, we should describe the property of the rational number field

$$\text{Gal}(\bar{\mathbb{Q}}/\mathbb{Q})$$

More genrally

Definition 0.0.0.1 Let G be a group, then G is **relizable** over K if there exists some L such that

$$\text{Gal}(L/K) = G$$